

PowMon

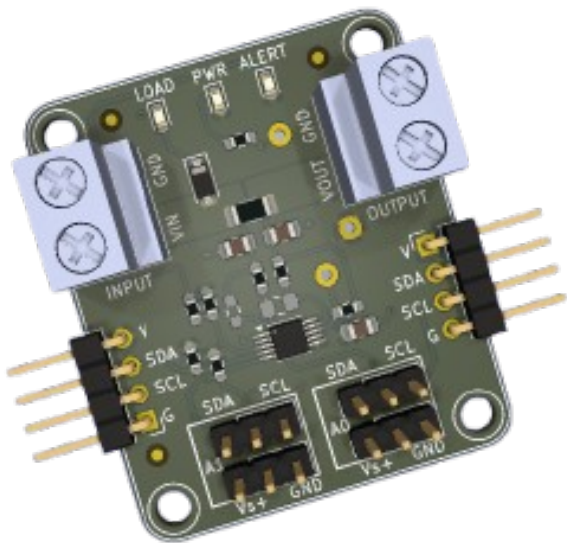
Variant: No Variant

2025-04-01
Rev V1

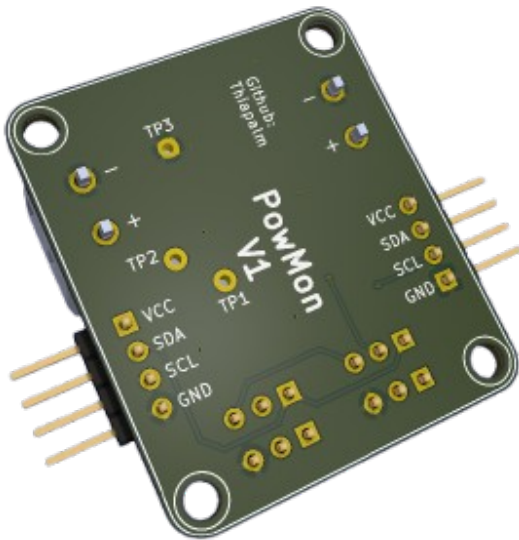
RELEASED

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11		21		31	
2	Block Diagram	12		22		32	
3	Project Architecture	13		23		33	
4	Sense	14		24		34	
5	Revision History	15		25		35	
6		16		26		36	
7		17		27		37	
8		18		28		38	
9		19		29		39	
10		20		30		40	

TOP VIEW



BOTTOM VIEW



NOTES

Add a comment here

Not fitted components are marked as **X**

- DRAFT - Very early stage of schematic, ignore details.
- PRELIMINARY - Close to final schematic.
- CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
- RELEASED - A board with this schematic has been sent to production.

No Variant 2025-04-01

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

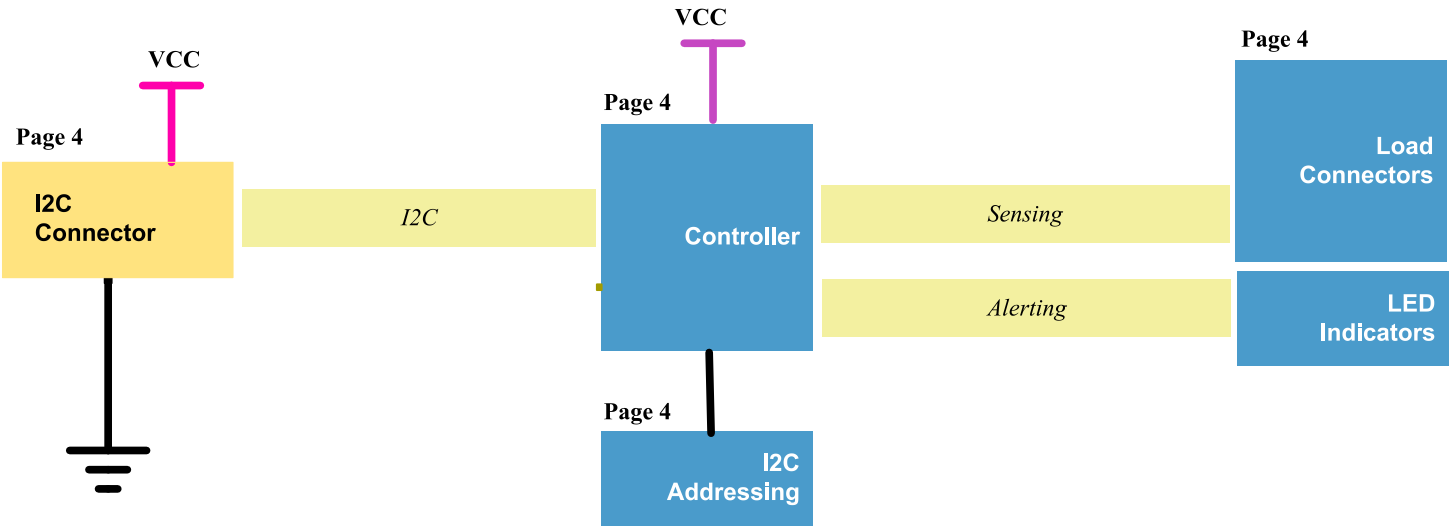
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

	Comments:		Company: thiago.palmieri@adastra.alceas.com		Variant: No Variant	
	Board Name: PowMon		Project Name: PowMon			
	Sheet Title: Cover Page	File Name: KDT_Hierarchical.kicad_sch	Designer: Thiago Palmieri	Date: Last Modified Date	Revision: V1	
Sheet Path: /			Reviewer: Thiago Palmieri	Size: A3	Sheet: 1 of 5	

[2] Block Diagram



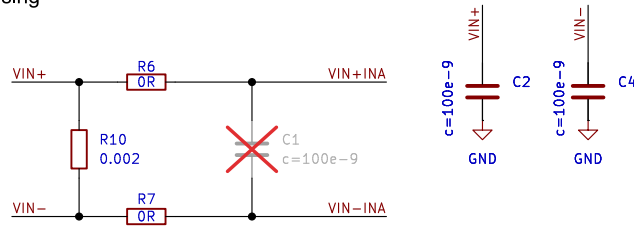
Target specifications:

Input voltage:	2.7 - 5.5 V
BUS Voltage	0 - 36V
ADC Conversion Time	140us - 9ms

	Comments:	Company: thiago.palmieri@adastra.alceas.com		Variant: No Variant	
		Board Name: PowMon		Project Name: PowMon	
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Thiago Palmieri	Date: Last Modified Date	Revision: V1
	Sheet Path: /Block Diagram/		Reviewer: Thiago Palmieri	Size: A3	Sheet: 2 of 5

</

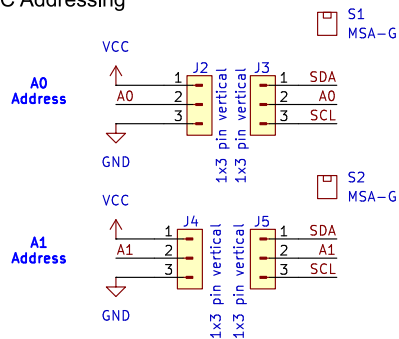
DESIGN NOTE: Sensing



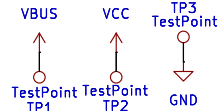
DESIGN NOTE: Load Connectors



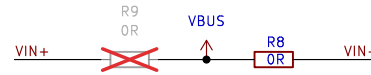
DESIGN NOTE: I2C Addressing



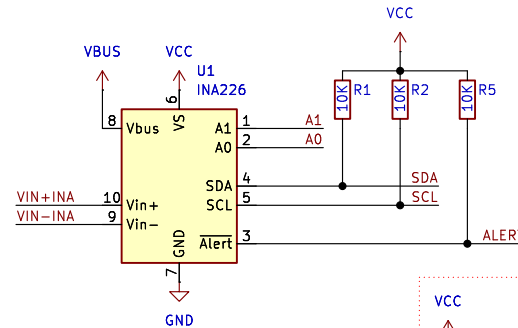
[4] Sense



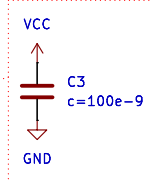
DESIGN NOTE: VBUS Ref



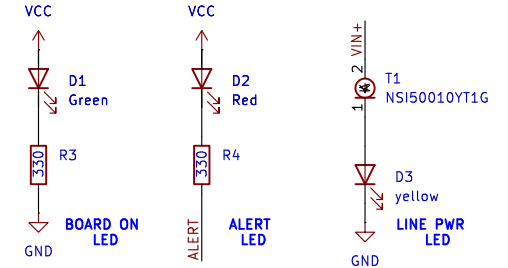
DESIGN NOTE: Controller



DESIGN NOTE: Close to VS pin



DESIGN NOTE: LED Indicators



DESIGN NOTE: I2C Connector

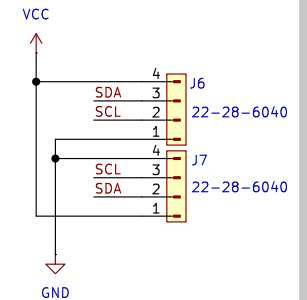


Table 6-2. Address Pins and Slave Addresses

A1	A0	SLAVE ADDRESS
GND	GND	1000000
GND	VS	1000001
GND	SDA	1000010
GND	SCL	1000011
VS	GND	1000100
VS	VS	1000101
VS	SDA	1000110
VS	SCL	1000111
SDA	GND	1001000
SDA	VS	1001001
SDA	SDA	1001010
SDA	SCL	1001011
SCL	GND	1001100
SCL	VS	1001101
SCL	SDA	1001110
SCL	SCL	1001111

Comments:

AN5346
STM32G474 Datasheet p.81
J. Pieper ADC investigation

Sheet Title:

Sense

Sheet Path:

/Project Architecture/Sense/

Company:

thiago.palmieri@adastra.aleas.com

Board Name:

PowMon

File Name:

Sense.kicad_sch

Designer:

Thiago Palmieri

Reviewer:

Thiago Palmieri

Variant:

No Variant

Project Name:

PowMon

Date:

Last Modified Date

Size:

A4

Revision:

V1

Sheet:

4 of **5**

[5] Revision History

01.04.2025 - V1 Revision
Variant: No Variant

- Board Creation on Kicad
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-

DD.MM.YYYY - xxx Revision
Variant: xxx

-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-

DD.MM.YYYY - xxx Revision
Variant: xxx

-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-

DD.MM.YYYY - xxx Revision
Variant: xxx

-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-
-

	Comments:	Company: thiago.palmieri@adastra.aleeas.com		Variant: No Variant	
		Board Name: PowMon		Project Name: PowMon	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Thiago Palmieri	Date: Last Modified Date	Revision: V1
	Sheet Path: /Revision History/		Reviewer: Thiago Palmieri	Size: A4	Sheet: 5 of 5